ADITYA UNIVERSITY

Department of Artificial Intelligence And Machine Learning

Web Application Development Using Mernstack

Cornerstone Project Final Review

Employee Record Management System (MERN CRUD)

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Presentation Outline

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ABSTRACT

The Employee Record Management System (MERN Stack) is a full-stack web application designed to efficiently manage employee data in a centralized and organized manner. The system is developed using the MERN stack (MongoDB, Express.js, Node.js, and frontend technologies) to provide a scalable and dynamic solution for handling employee records.

The application enables administrators to perform complete CRUD (Create, Read, Update, Delete) operations on employee data, including details such as name, email, department, and salary. It also incorporates advanced features like searching employees by name and filtering based on department and salary, which enhances data retrieval and usability.

The backend is implemented using Node.js and Express.js to create RESTful APIs, while MongoDB is used for efficient data storage and management. The frontend is designed to provide an intuitive and user-friendly interface for seamless interaction.

PROJECT OVERVIEW

PROBLEM STATEMENT

Organizations require a centralized system to manage employee records efficiently. Manual methods lead to poor data organization, time-consuming searches, and salary inconsistencies.

The project focuses on developing a MERN-based web application that performs CRUD operations and enables department and salary-based search filtering.

OBJECTIVES

To develop a web-based Employee Record Management System using the MERN stack that enables efficient management of employee details, salary information, and department data through CRUD operations and search filtering.

INTRODUCTION

- The Employee Record Management System (MERN Stack) is a web-based application developed to simplify and automate the management of employee information within an organization. In many companies, employee data such as personal details, department, and salary are often maintained manually or in scattered systems, leading to inefficiency, data redundancy, and difficulty in accessing information.

- To overcome these challenges, this project provides a centralized digital solution using the MERN stack (MongoDB, Express.js, Node.js, and frontend technologies). The system allows administrators to efficiently perform CRUD operations—adding, viewing, updating, and deleting employee records in a structured manner.

- In addition, the application includes search and filtering features, enabling users to quickly find employees based on name, department, or salary. This improves data accessibility and helps in better decision-making.

Analysis & Design

System Users

- **Admin**

Functional Requirements

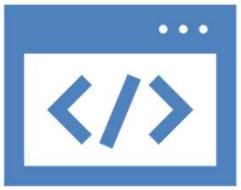
- Add Employee
- View Employees
- Update Employee
- Delete Employee
- Search by Name
- Filter by Department
- Filter by Salary

Employee Schema (MongoDB)

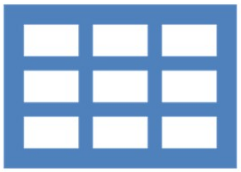
- _id
- name
- email
- department
- salary
- username
- password
- position
- status

Note: Data is stored in MongoDB using a schema defined with Mongoose.

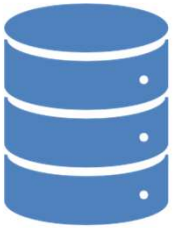
SYSTEM ARCHITECTURE



Frontend (React.js)



Backend (Express.js REST API (Node.js))



MongoDB Database

TECHNOLOGY STACK

Frontend Implementation.

Frontend: React.js **& Styling :** CSS

Used **JavaScript** for search and filter functionality.

Backend Implementation.

Node.js as Runtime Environment and **Express.js** for REST API development

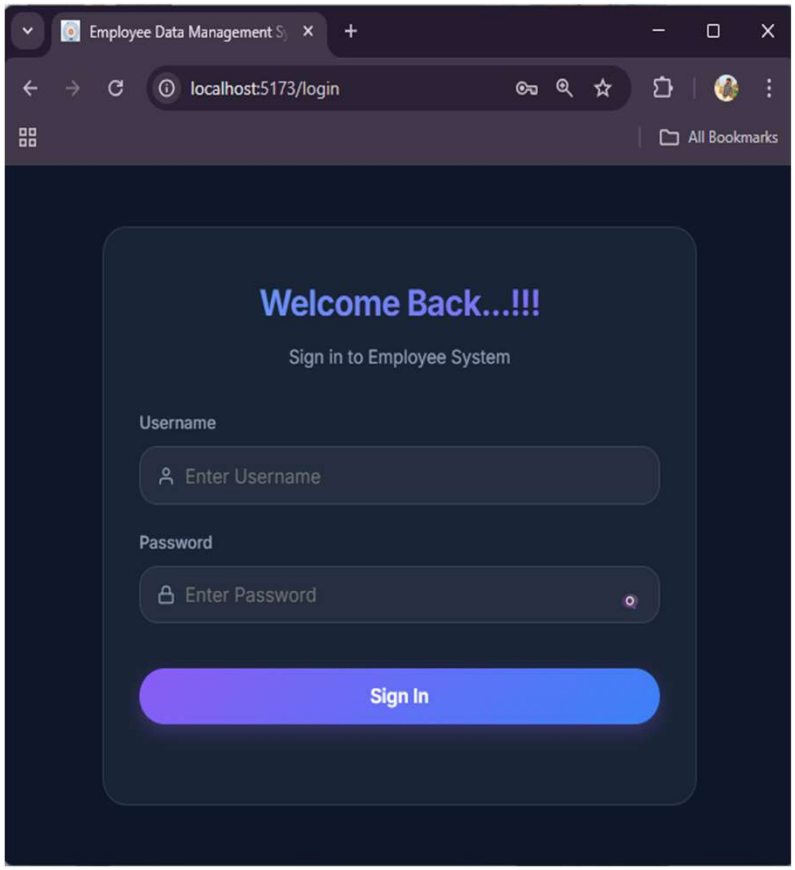
Database Implementation.

MongoDB for Database Storage

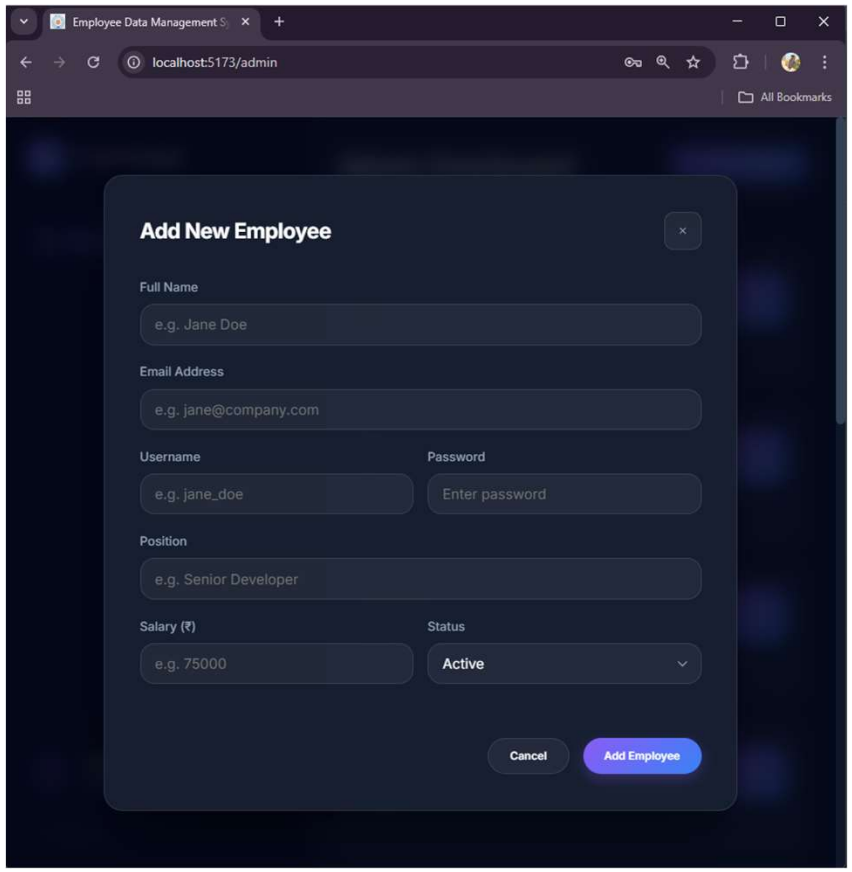
Mongoose for Schema & Database Integration.

OUTPUT SCREENS

Output Screens

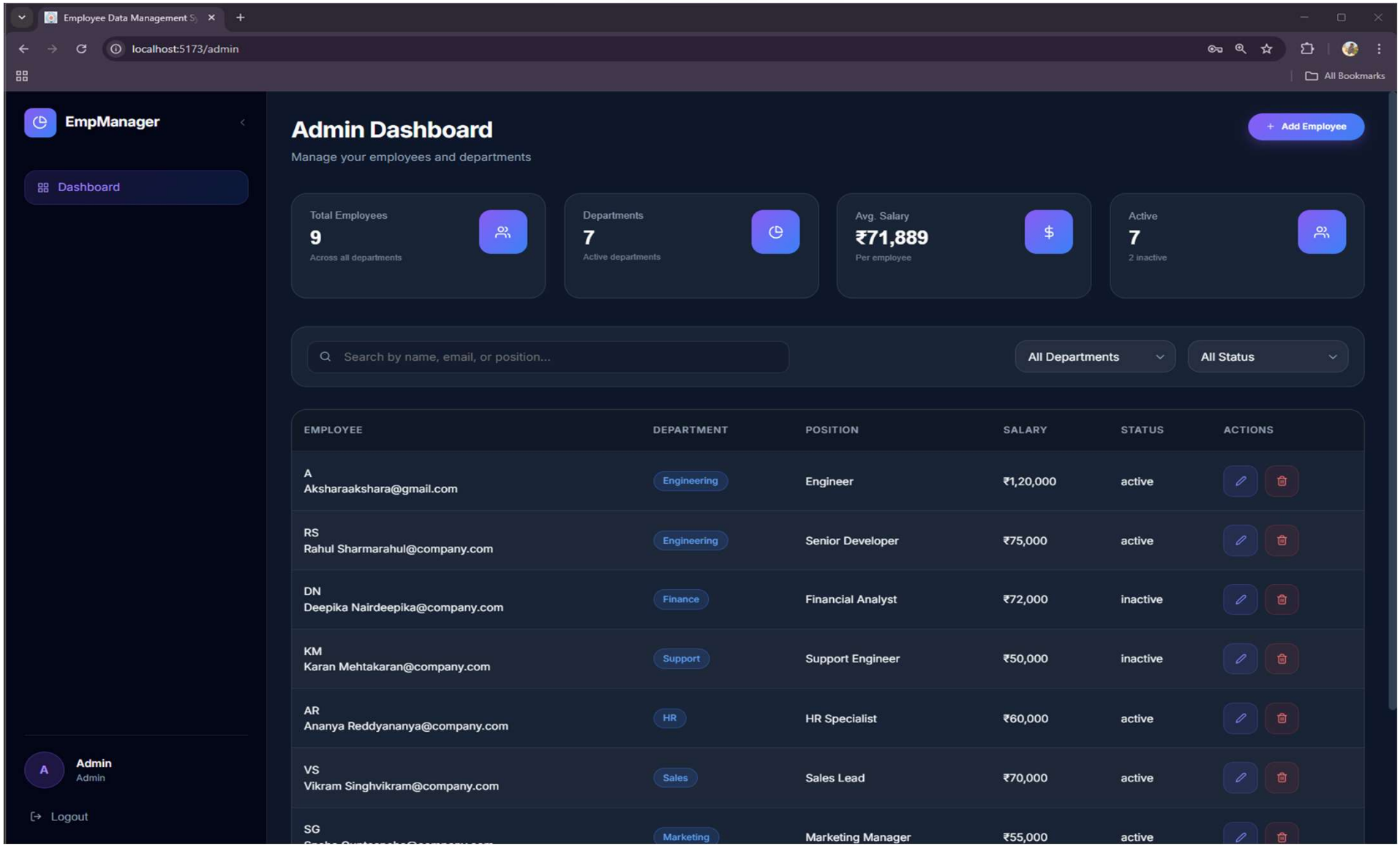


Login Page



Creation of New Employee to the company

OUTPUT SCREENS



Admin Dashboard

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CONCLUSION

- The Employee Record Management System successfully provides a centralized platform for managing employee details, salary information, and department records.
- The system implements complete CRUD functionality along with search and filtering features using the MERN stack.
- Through this project, we demonstrated full-stack development by integrating frontend, backend, and database technologies to build a functional and scalable web application.

THANK YOU